GENERAL SPECIFICATION G4.7

# Sampling Systems (Used Water)

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## G4.7 SAMPLING SYSTEMS (USED WATER)

### G4.7.1 General Requirements for Sampling

(a) Sample points: Sample points shall be provided as specified to provide a means of water quality monitoring, efficiency monitoring, controlling the works and to allow comparison of performance of different process units and other purposes where recommended by the Contractor to suit his design.

(b) Location

(i) Sample points shall be positioned in a manner which ensures that the sample is properly representative of the fluid stream from which it is taken in terms of physical, chemical and microbiological parameters.

(ii) Special care shall be taken following chemical dosing to ensure that mixing has taken place sufficiently before the sample is withdrawn.

(iii) In pipelines up to 600 mm in diameter, the limiting position of the sample point shall be where the ratios of minimum concentration to maximum concentration of the dosed chemical across the horizontal and vertical diameters of the pipeline are at least 0.95.

(iv) In pipelines larger than 600 mm in diameter, these ratios of concentrations shall not be less than 0.90 or, where applicable, such higher value as may be necessary to achieve the stated plant performance guarantee. Similar criteria shall apply to flow in channels and rectangular ducts of all sizes except that the concentrations shall be across the horizontal and vertical centre axes.

(c) Take-Off Arrangement

(i) Under no circumstances shall samples be withdrawn directly from the pipe or channel wall. Sample withdrawal tubes shall project into the bulk fluid flow in a manner which avoids vibration and the collection of entrained gas bubbles or sediments.

(ii) Sample tapping points shall not be less than 25 NB. Generally, any reduction in pipe size shall take place after the first isolating valve.

(iii) The take-off arrangement up to and including the first isolating valve shall be designed for maximum mechanical strength in addition to hydraulic considerations and the integrity of the main fluid stream shall not be prejudiced by any failure in or damage to the sampling system itself.

(iv) Samples from pipes for closed loop control shall be withdrawn from the central 1/3$^{rd}$ of the pipe by means of withdrawal tubes or lances. The sample pipe shall have substantial support shaft to prevent velocity induced stress and vortex shedding and a profile head to minimise hydraulic interference effects.

(v) Samples drawn from pipes downstream of mixers shall be provided with a flanged seal housing to permit insertion and withdrawal of the sample probe under pressurised pipeline condition.

(d) Sampling Rate

(i) The sampling rate shall be optimised for each sampling system with due regard to the following:

1) The purpose of sampling;

2) Maintenance of sample quality;

3) Effective monitoring and control, where applicable;

4) Hydraulic restraints; and

5) Minimisation of wastage.

(ii) The sample delivery rate to analysers shall be as recommended by the instrument manufacturers and any necessary pressure or flow regulating device to maintain this shall be provided.

(iii) The sample delivery rate to sample taps shall be limited to 2 litres per minute per tap by means of suitable pre-set or flow limiting valves but not by use of isolating valves.

(iv) Generally, samples shall be delivered at rates significantly more than those required by analysers and sample taps to satisfy all relevant design criteria. Sample pipework shall not be less than 15 mm nominal bore (NB) diameter under all circumstances and, where hydraulic or water quality conditions dictate, shall be larger.

(e) Sample Recovery and Disposal

(i) Unless otherwise specified, excess sample water shall be recovered for re-use if there is no risk of contamination and it shall be returned to the process at a suitable point, subject to the S.O.'s approval.

(ii) In this way, wastage shall generally be limited to discharge of used samples from analysers and from sample taps.

(iii) All necessary means shall be provided for collection and disposal of waste sample water with due regard to any contaminants which may be introduced within the sampling facility such as analytical reagents or cleaning compounds.

(f) Sample Validation

(i) All sample systems shall be provided with pressure gauges and the gauge diameter shall be 100 mm. The gauges shall be mounted at the receiving end of the sample system such as above the sample sink.

(ii) Where samples are delivered to water quality analysers, that part of the flow received by each analyser shall be validated by means of a flow switch or, where hydraulic conditions permit, a pressure switch. In either case, volt-free contacts shall be provided for initiation of an alarm condition and, where applicable, instrument and/or reagent shut-down.

### G4.7.2 Sample Pumps

(a) Unless otherwise specified or approved, sample pumps shall be of the horizontal end-suction back pull-out centrifugal, vertical in-line centrifugal or progressive cavity type with flange mounted three phase cage induction electric motors. The final selection of the sample pump type shall take consideration of ensuring consistent and stable flow delivery against the pipeline system head to the designated locations.

(b) The pumps shall have close grained cast iron bodies, stainless steel rotors, synthetic nitrile rubber stators and mechanical shaft seals.

(c) Sample pumps shall be installed above flood level wherever possible and shall be readily accessible for maintenance purposes.

(d) Over-pressure protection shall be provided either by means of an integral device or a separate pressure relief valve.

(e) Additional requirements on pumps shall be according to the General Specification on Pumps.

(f) Where specified, sample pumps shall be installed in duplicate and arranged for automatic changeover on failure. Otherwise, at least one (1) spare unit of sample pump shall be provided and stored at the designated store room for every four (4) installed pump units. Single pumps shall be powered by way of a local plug and socket arrangement to facilitate easy substitution.

(g) The sample analysers shall need to be stored in appropriately designed housing / cabinet located outside of hazardous zones, be rated to a minimum of IP55 and subject to S.O.'s approval.

### G4.7.3 Sample Pipework

(a) General - Sample pipework shall be selected to suit the application and type of measurement from either uPVC, stainless steel or polyethylene. In addition, medium density polyethylene pipe may be used where appropriate only for pipe sections run through buried ducts or for pipe sections laid directly in the ground.

(b) uPVC Pipework

(i) uPVC sample water pipework shall be the `hi-impact' type to BS 3505 Class E with solvent welded joints and fittings complying with the relevant parts of BS EN 1452.

(ii) Screwed connections to metal components such as pressure gauges, etc. shall be made with proprietary iron to uPVC adaptor fittings.

(iii) Pipework shall be supported in accordance with manufacturers' recommendations and adequate provision shall be made for thermal expansion and contraction.

(iv) uPVC pipework shall not be used where it may influence analytical measurements such as dissolved oxygen concentration.

(c) Stainless Steel Pipework

(i) Stainless steel sample water pipework shall be type 316 S11 or higher grade to BS EN 10216-5 with tolerances in outside diameter to Table 1 of BS EN ISO 8434 Parts 1 to 4 and tested in accordance with Category 2 requirements.

(ii) Pipe couplings shall be light series compression Type A to BS EN ISO 8434 Parts 1 to 4.

(iii) Pipework shall be supported directly on surfaces with proprietary clips or, where several pipelines follow a parallel route, they may be clipped to trays dedicated to this service.

(d) Polyethylene Pipework

(i) Polyethylene sample water pipework in accordance with BS EN 12201 may be used for those pipe sections which are run through buried ducts or laid directly in the ground.

(ii) Pipe couplings shall be of the electro-fusion type and incorporate pins which pop up to confirm that optimum melt pressure has been achieved.

### G4.7.4 Sample Isolating Valves

(a) General - The valves shall be compatible with the main sample pipework system and be of uPVC or stainless steel material.

(b) uPVC valves - Isolating valves shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms. Assembly nuts and bolts shall be stainless steel.

(c) Stainless steel valves - Isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in stainless steel 316C16 to BS EN 10293, PTFE diaphragms and top works in standard proprietary materials. Assembly nuts and bolts shall be stainless steel. The valves shall be tested in accordance with BS EN 10293.

### G4.7.5 Sample Taps

(a) Sample taps shall be in chrome plated brass, 15 NB BSP size, with cross handle and outlet nozzle suitable for sterilisation by flaming. The nozzles shall have protective screw on caps which are attached by means of chain to the tap spindle.

### G4.7.6 Sample Sinks

(a) Sample sinks shall be to BS 1206, of 455 mm nominal width, 255 mm nominal depth and the following nominal lengths to suit the number of sample taps:

(i) Up to 4 samples: 610 mm;

(ii) 5 samples: 760 mm; and

(iii) 6 samples: 915 mm.

(b) The sinks shall generally be supported from a wall on cantilever brackets but where this is impracticable, they may be supported on stanchions from the floor. The base of the sink shall be approximately 700 mm from the floor and shall be fitted with a drain connected to a designated location approved by the S.O.. There shall be provision such as shelves, tables, etc. for placement of sampling bottles, tools, cleaning items and all associated paraphernalia.

(c) The sample tap nozzles shall be positioned at 100 mm centres at a distance of 170 mm from the rear of the sink and at a suitable height above the sink to allow collection of samples in a glass bottle approximately 300 mm high.

(d) Sample sinks shall be equipped with all necessary potable water supply piping, bib taps, valves and flushing hose of washing and cleaning purposes.

**G4.7.7 Automatic Wastewater Samplers**

(a) Sampler Unit

(i) Unless otherwise specified, the automatic sampler shall be refrigerated and compartmentalised type with drainage to a designated location approved by the S.O.. Sampler shall be heavy-duty, suitable for outdoor installation, automatic and capable of sampling wastewater treatment plant process streams;

(ii) Each sampler shall be suitable for the solids size within the samples as following:

1) 6 mm maximum for wastewater; and

2) 10 mm for sludge.

(iii) Samples shall be taken from a continuous flow of process stream supplied from an external source under process pressure condition except for the sampler at the influent channel which is at atmospheric pressure;

(iv) Each sampler shall be line-fed type;

(v) Sampler shall operate by tapping into the flow and drawing off a measured sample;

(vi) A stainless-steel sampling reservoir and sampler funnel shall be furnished as part of each sampler and shall be capable of being isolated from the main portion of the sampler to facilitate cleaning;

(vii) The sampling reservoir shall be designed to receive a sample flow rate high enough to provide sufficient velocity and turbulence to prevent sample separation and enable the taking of a truly representative sample. Each chamber shall have inlet and outlet or drain connections;

(viii) Each sampler shall consist of an air-operated sampling dipper which is completely separated from the flow and mounted above the sampling reservoir. Cups shall be provided with the dipper mechanism to obtain samples of 500 or 400 ml volumes;

(ix) The sampler storage compartment shall be refrigerated, insulated and thermostatically controlled to maintain the temperature at 4-degree C to 10-degree C. Fan-cooled condenser shall be used unless otherwise specified. Exterior shall be all stainless steel;

(x) The refrigerator capacity of the sampler unit shall be capable for 24 sample bottles of one (1) litre each. The sample volume shall be from 200 mL to 1 L;

(xi) The sampler shall be designed with the flexibility of storing sample collection in a single bottle or groups of bottles and changing over between bottles without the use of special tools.

(xii) Supply each sampler with distribution tray for 24 sets of sample containers, 1 litre capacity each and 4 nos. of spare bottles per set of containers;

(xiii) The sampler dipper, reservoir, and funnel shall all be mounted together in a neat, compact, stainless steel cabinet with a sample storage compartment below. The cabinet shall be equipped with a hinged door to provide access to the storage compartment;

(xiv) Each sampler shall be suitable for mounting on level concrete base with a spacing of 100 mm between the rear of its cabinet and the wall backing it;

(xv) Sample wetted material shall be PVC;

(xvi) The sample unit shall be operated on 230-volt, 50-Hz, single-phase power; and

(xvii) Enclosure

1) Constructed of shock-proof Polyethylene (PE) with integrated lockable carrying handles to prevent unauthorised access.

2) Covering entire sampler unit, full front opening door with integrated lock and stainless steel full-length hinge;

3) Suitable for mounting on level concrete base with 10 mm minimum diameter stainless steel anchor bolts;

4) Enclosure shall be separate from sampler cabinet and shall be easily removed for maintenance; and

5) Provided with protective splash cover.

(b) Sampler Operation for Wastewater

(i) The sampler dipper shall operate automatically either based on internally programmed time intervals or flow-paced based on a remote electronic flow signal. The electrical signal shall operate an air solenoid valve to provide compressed air for the operation of the dipper mechanism. The air solenoid actuator valve, supplied by the sampler manufacturer, shall be mounted at the top of the dipper drive air cylinder. The sampler manufacturer shall provide an air filter, pressure regulator and gauge to each sampler.

(ii) Each sample taken by the dipper mechanism shall be automatically poured into the funnel and conveyed first through a stainless-steel tube and then a plastic tube into a bottle in the storage compartment below.

(c) Sampler Operation for Sludge

(i) Sampler shall operate with a positive pressure exerted at the sample tapping point by the sludge being sampled. When subjected to this pressure, the sludge will flow freely along the sample pipe.

(ii) High-pressure air purge shall flush the sampler inlet and connection pipe. A volume of the sludge being sampled is introduced into the sample chamber. A second high pressure air purge shall flush excess sludge from the sample chamber back to its source, thus leaving a residual volume in the sample chamber. The inlet valve closes and the pressure in the chamber is reduced. The residual volume (which constitutes the sample) shall be exhausted under pressure from the sample chamber through the outlet pipe and into the sample collection bottle.

(d) Sample Control for both Waste Water and Sludge

(i) The sampler shall be capable of sampling a predetermined fraction of the process flow by either constant rate sampling based on programmed time or flow paced by remote flow signal. It shall be capable of accepting remote initiation via volt-free contact closure or 4 -20 mA proportional signal from the flowmeter.

(ii) The sampler shall be furnished complete with an integral Control Panel. Panel shall have the following:

1) Test button;

2) 'Bottle Full' warning light;

3) Operator interface with LCD display for programming; and

4) Contact outputs for sample collected, plugged intake, purge failure etc.

(iii) The control panel of the sampler shall have a precision timer operated in the range of 0.01 to 999 minutes.

**END OF SECTION**